

Notes: Djankov, La Porta, Lopez-de-Silanes, and Shleifer (QJE, 2002)

The Regulation of Entry

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1 Big picture

- **Question.** Why do countries differ so much in the number of official procedures, time, and cost required to start a legal business?
- **Core contribution.** The paper creates a new cross-country dataset on formal entry regulation for start-up firms in 85 countries and uses it to test public-interest versus public-choice theories of regulation.
- **Main empirical fact.** Entry regulation is extremely burdensome in many countries. The number of procedures ranges from 2 to 21, the official time ranges from 2 to 152 business days, and official cost ranges from less than 0.5% of per capita GDP to more than 4.6 times per capita GDP.
- **Theories tested.** Public-interest theory says entry regulation screens bad firms and protects consumers. Public-choice theory says regulation creates rents for incumbent firms, politicians, and bureaucrats.
- **Main evidence.** Heavier entry regulation is associated with more corruption and a larger unofficial economy, but not with better product quality, less pollution, better health outcomes, or greater competition.
- **Political economy result.** Less democratic and less limited governments regulate entry more heavily, even controlling for income. This supports the tollbooth/public-choice view.
- **Takeaway.** Entry regulation often appears to be a barrier that creates rents rather than a tool that corrects market failures.

2 Incremental contribution of the paper

- **New measurement contribution.** Before this paper, systematic cross-country evidence on the exact procedures, official time, and official cost of business entry was limited. The paper turns legal entry barriers into comparable quantitative variables for 85 countries.
- **Theory-testing contribution.** It does not merely document red tape. It uses the data to distinguish public-interest regulation from public-choice regulation by examining both the consequences of regulation and the political attributes of regulators.
- **Public-choice evidence contribution.** The paper provides early large-sample evidence that formal regulation is positively related to corruption and informal activity, supporting the idea that regulation can be a tool for rent extraction.
- **Political institutions contribution.** It connects product-market entry barriers to constitutional and political variables - executive constraints, political rights, autocracy, legal origin - thereby integrating entry regulation into comparative institutional economics.
- **Policy contribution.** The findings provide a foundation for business-entry reform agendas, including what later became the World Bank Doing Business “Starting a Business” indicators.
- **Methodological contribution.** The paper emphasizes official requirements, official costs, and official time rather than unofficial bribe costs. This makes the database replicable and conservative: corruption and bureaucratic delays would only add to the burden.

3 Data and measurement

3.1 Sample and hypothetical firm

The dataset covers 85 countries in 1999. The authors study the official procedures required for a standardized start-up to legally begin operations. The hypothetical firm is designed to be simple and comparable across countries: a domestically owned start-up engaged in ordinary industrial or commercial activity, with no foreign ownership, no special environmental hazards, and no need for sector-specific licenses.

Interpretation: The standardized firm is essential for cross-country comparability. Without it, differences in procedures could reflect differences in firm size, industry, ownership, or legal form rather than the regulatory system itself.

3.2 Data collection process

The authors collect information from written laws, government publications, official web pages, development-agency reports, and local law firms. They then cross-check discrepancies with government agencies and legal practitioners. The data record official requirements only.

- **Number of procedures:** all official steps the entrepreneur must complete before legal operation.
- **Time:** minimum official business days needed to complete procedures without delays.
- **Cost:** official monetary cost of completing the procedures, scaled by per capita GDP.
- **Cost plus time:** cost plus the monetized value of time, using GDP per capita to value days lost.

Interpretation: The official-cost focus makes the paper's estimates conservative. The authors explicitly do not include bribes or bureaucratic delays; therefore, the real burden in corrupt countries is likely larger than measured.

3.3 Procedure categories

Procedures are grouped into five categories:

1. screening procedures;
2. tax-related requirements;
3. labor/social-security-related requirements;
4. safety and health requirements;
5. environment-related requirements.

Interpretation: Screening is a residual category that includes approvals, notarizations, name checks, and certificates whose stated purpose is not obviously tied to health, tax, labor, or environmental compliance. The prevalence of screening is important because it is most naturally interpreted as an entry barrier or tollbooth.

3.4 Outcome variables used to test theories

The paper studies two classes of outcomes.

Public-interest outcomes:

- ISO 9000 certifications as a proxy for product-quality standards;
- water pollution;
- deaths from accidental poisoning;
- deaths from intestinal infection;
- product-market competition.

Public-choice outcomes:

- size of the unofficial economy as a share of GDP;

- employment in the unofficial economy;
- corruption index.

Interpretation: Public-interest theory predicts that entry regulation should be associated with better product quality, fewer externalities, and more orderly markets. Public-choice theory predicts more corruption, a larger informal sector, and weaker competition.

3.5 Political and legal-origin variables

The political variables measure whether government is democratic, limited, and accountable:

- de facto independence of the executive;
- constraints on executive power;
- effectiveness of the legislature;
- competition in the nominating process;
- autocracy;
- political rights.

Legal origins are classified as English, French, German, Scandinavian, or Socialist.

Interpretation: These variables test which governments regulate entry. Public-interest theory predicts more representative governments should regulate more if regulation protects citizens. Public-choice theory predicts the opposite: less constrained governments should regulate more because they can extract more rents.

4 Empirical specifications

4.1 Cross-country outcome regressions

The main consequences-of-regulation regressions take the form

$$Y_c = \alpha + \beta \log(\text{Procedures}_c) + \gamma \log(\text{GDPpc}_c) + \varepsilon_c,$$

where Y_c is a social outcome such as corruption, unofficial economy, pollution, health outcome, or product-market competition.

Interpretation: The coefficient β asks whether countries with heavier entry regulation have better public-interest outcomes or worse public-choice outcomes, optionally controlling for income.

4.2 Tollbooth regressions

For corruption, the paper estimates

$$\text{Corruption}_c = \alpha + \beta_1 \log(\text{Procedures}_c) + \beta_2 \log(\text{GDPpc}_c) + \varepsilon_c,$$

and analogous regressions using log time and log cost.

Interpretation: A negative corruption-index coefficient means that more procedures are associated with worse corruption because the corruption index is scaled so that lower values indicate more corruption. Strong negative coefficients support the tollbooth view.

4.3 Political institutions regressions

To test which governments regulate, the paper estimates

$$\log(Procedures_c) = \alpha + \delta PoliticalInstitution_c + \eta \log(GDPpc_c) + \varepsilon_c.$$

The political variables include executive constraints, legislative effectiveness, political rights, autocracy, and legal origin.

Interpretation: If public-interest theory were correct, more democratic and accountable governments should impose more beneficial regulation. Instead, the paper finds that less accountable governments impose more procedures.

4.4 Income-split and robustness logic

The paper also splits countries by above/below median GDP per capita and re-estimates corruption regressions. It further checks whether press freedom and judicial-access variables could explain the absence of social benefits from regulation.

Interpretation: These exercises respond to the argument that poor countries have greater market failures or weaker alternatives to regulation. The main results do not support that interpretation.

5 Identification strategy

5.1 What the paper can and cannot identify

This is a cross-country observational paper. It does not randomly assign entry regulation. The empirical strategy is comparative institutional evidence rather than a causal natural experiment.

Interpretation: The paper's identification is theory-discriminating rather than design-based causal identification. It asks which broad theory best fits cross-country patterns in regulation, corruption, informality, social outcomes, and political institutions.

5.2 Key identifying comparisons

- **Consequences comparison:** Public-interest theory predicts socially better outcomes where regulation is heavier; public-choice theory predicts corruption and informality.
- **Political-institutions comparison:** Public-interest theory predicts more regulation by democracies and accountable governments; public-choice theory predicts more regulation by less constrained governments.
- **Legal-origin comparison:** Civil-law and socialist origins proxy for a stronger state and greater interventionism; English origin proxies for lighter regulatory tradition.
- **Income controls:** Including GDP per capita helps separate regulation from general development and state capacity.

Interpretation: The strongest evidence for the paper comes from convergence across tests: regulation is linked to corruption and informality, not public-good improvement, and it is heavier under less democratic and more interventionist governments.

5.3 Main threats

- Countries with heavier regulation may have more market failures that are only imperfectly measured.
- Income and institutional quality are highly correlated, making causal interpretation difficult.
- Official rules may differ from actual enforcement; informal costs and delays are not measured.
- Measures of product quality, pollution, competition, and corruption are cross-country proxies rather than direct measurements of entrant quality or regulatory abuse.

Interpretation: The paper addresses these concerns by using multiple outcomes, multiple regulation measures, and multiple political variables. Still, the estimates should be read as strong descriptive institutional evidence rather than experimental causal effects.

6 Tables and figures

6.1 Table I: List of Procedures for Starting up a Company and Figures I–II: Start-up Procedures in New Zealand and France

Read:

- Table I lists common start-up procedures across countries, including screening, tax, labor/social-security, health/safety, and environmental requirements.
- Figure I shows the streamlined New Zealand process: only three procedures, very little time, and minimal cost.
- Figure II shows the France process: fifteen procedures, much longer time, and sharply higher cumulative cost.

Interpretation: Table I makes the content of regulation tangible, while Figures I and II show how different institutional systems translate into procedure sequences. New Zealand looks like a registration system; France looks like a multilayer authorization system.

Economic message: The comparison illustrates the core institutional variation: entry regulation can be a short notification process or a long chain of approvals and fees.

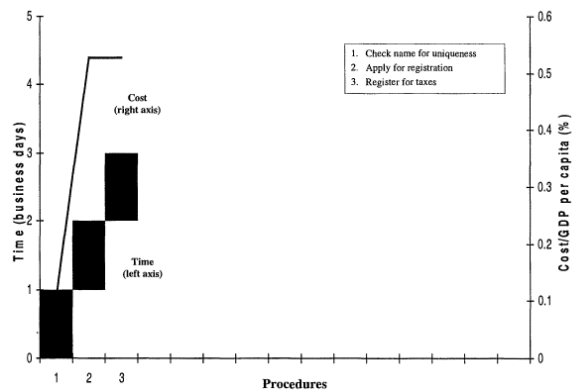


FIGURE I
Start-up Procedures in New Zealand

Procedures are lined up sequentially on the horizontal axis and described in the text box. The time required to complete each procedure is described by the height of the bar and measured against the left scale. Cumulative costs (as a percentage per capita GDP) are plotted using a line and measured against the right scale.

(a) Figure I: Start-up procedures in New Zealand.

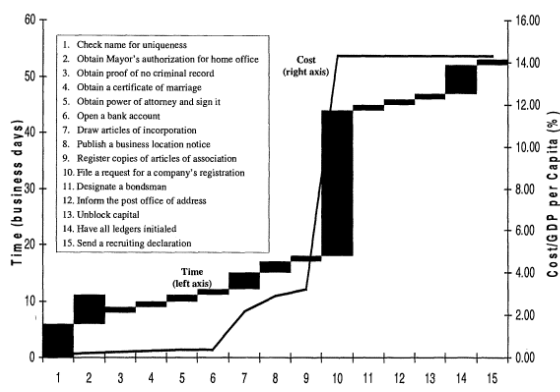


FIGURE II
Start-up Procedures in France

Procedures are lined up sequentially on the horizontal axis and described in the text box. The time required to complete each procedure is described by the height of the bar and measured against the left scale. Cumulative costs (as a percentage per capita GDP) are plotted using a line and measured against the right scale.

(b) Figure II: Start-up procedures in France.

TABLE I
LIST OF PROCEDURES FOR STARTING UP A COMPANY
This table provides a list of common procedures required to start up a company in the 85 countries of the sample.

1. Screening procedures
- Certify business competence
- Certify a clean criminal record
- Certify marital status
- Check the name for uniqueness
- Notarize company deeds
- Notarize registration certificate
- File with the Statistical Bureau
- File with the Ministry of Industry and Trade, Ministry of the Economy, or the respective ministries by line of business
- Notify municipality of start-up date
- Obtain certificate of compliance with the company law
- Obtain business license (operations permit)
- Obtain permit to play music to the public (irrespective of line of business)
- Open a bank account and deposit start-up capital
- Perform an official audit at start-up
- Publish notice of company foundation
- Register at the Companies Registry
- Sign up for membership in the Chamber of Commerce or Industry or the Regional Trade Association
2. Tax-related requirements
- Arrange automatic withdrawal of the employees' income tax from the company payroll funds
- Designate a bondsman for tax purposes
- File with the Ministry of Finance
- Issue notice of start of activity to the Tax Authorities
- Register for corporate income tax
- Register for VAT
- Register for state taxes
- Register the company bylaws with the Tax Authorities
- Seal, validate, rubricate accounting books
3. Labor/social security-related requirements
- File with the Ministry of Labor
- Issue employment declarations for all employees
- Notarize the labor contract
- Pass inspections by social security officials
- Register for accident and labor risk insurance
- Register for health and medical insurance
- Register with pension funds
- Register for Social Security
- Register for unemployment insurance
- Register with the housing fund
4. Safety and health requirements
- Notify the health and safety authorities and obtain authorization to operate from the Health Ministry
- Pass inspections and obtain certificates related to work safety, building, fire, sanitation, and hygiene
5. Environment-related requirements
- Issue environmental declaration
- Obtain environment certificate
- Obtain sewer approval
- Obtain zoning approval

Table I: list of procedures for starting up a company.

6.2 Table II: Variables and Table III: The Data

Read:

- Table II defines all dependent and independent variables, including regulation measures, corruption, unofficial economy, public-good proxies, and political variables.
- Table III reports country-level data: number of procedures, procedure categories, time, cost, cost plus time, and GDP per capita.
- Countries are ranked from light regulation to heavy regulation.
- Canada, Australia, and New Zealand appear at the light end; countries such as the Dominican Republic, Madagascar, Mozambique, and Vietnam appear at the heavy end.

Interpretation: These tables are the empirical backbone of the paper. Table II clarifies variable construction; Table III shows the extraordinary dispersion in entry regulation across the 85-country sample.

Economic message: The data reveal that starting a legal business can be almost costless in some countries and prohibitively costly in others.

TABLE II
THE VARIABLES

This table describes the variables collected for the 85 countries included in our study. The first column gives the name of the variable. The second column describes the variable and provides the sources from which it was collected.

[illegible]

(a) Table II: variables, part 1.

TABLE III

Panel A reports the total number of procedures and their breakup in the following five categories: (1) safety and health; (2) environment; (3) taxes; (4) labor; and (5) screening. The table also reports the time, direct cost (as a fraction of GDP per capita in 1999) associated with meeting government requirements, and direct cost plus the monetized value of the entrepreneur's time (as a fraction of GDP per capita in 1999) as well as the level of GDP per capita in dollars in 1999. Countries are sorted in ascending order on the basis of (1) the total number of procedures; (2) time; and (3) cost. Panel B presents means of the variables by quartiles of GDP per capita in 1999. Panel C presents *t*-statistics for differences in means across quartiles of per capita GDP in 1999. Table 1 describes the variables in detail.

PROCEDURES TIME EVALUATION BY COUNTRY												
Country	Number of procedures		Safety & Health		Environment	Taxes	Labor	Sewerage	Time	Cost	Cost + time	GDP PPP
Point & Date												
Canada	2	0	0	1	0	0	1	2	0.0145	0.0225	19,320	19,320
Australia	2	0	0	1	0	1	0	2	0.0225	0.0305	20,050	20,050
United Kingdom	3	3	0	0	0	0	0	1	0.0172	0.0172	17,780	17,780
Denmark	3	0	0	1	0	0	2	3	0.1000	0.1120	10,520	10,520
Finland	3	0	0	1	0	0	0	1	0.1121	0.1197	10,140	10,140
United States	4	0	0	1	1	2	4	0.0049	0.0169	30,000	30,000	
Norway	4	0	0	0	1	2	18	0.0472	0.1302	12,280	12,280	
United Kingdom	5	1	0	0	0	0	1	0.0001	0.0001	24,640	24,640	
Hong Kong	5	0	0	0	1	4	15	0.0333	0.0903	12,330	12,330	
Mexico	5	0	0	0	1	0	0	0.0333	0.0444	12,111	12,111	
Malaysia	5	0	0	0	1	0	0	0.0116	0.1076	25,770	25,770	
Israel	5	0	0	0	2	1	2	0.2132	0.3412	15,860	15,860	
Spain	6	0	0	0	0	2	4	0.1285	0.1350	18,480	18,480	
Sweden	6	0	0	1	1	4	13	0.0256	0.0776	20,040	20,040	
Japan	6	0	0	0	2	1	3	0.1879	0.2879	2,100	2,100	
China	6	0	0	0	1	29	0.0049	0.1709	30	30	30	30
Peru	7	0	0	0	1	1	5	0.3074	0.3074	10,000	10,000	
Switzerland	7	0	0	0	1	2	0	0.1724	0.2064	16,360	16,360	
Singapore	7	0	0	0	2	1	4	0.1191	0.2071	27,610	27,610	
Latvia	7	0	0	0	1	2	3	0.4234	0.5154	4,470	4,470	
Poland	8	0	0	0	0	0	2	0.2655	0.4021	5,400	5,400	

(a) Table III: country data, part 1.

TABLE III

	(CONTINUED)										GDP PPPs
	Number of procedures	Safety & Health	Environment	Taxes	Trade	Foreign Labor	Screening	Cost	Cost + time	Cost	
Korea, Rep.	13	0	0	2	4	7	27	0.1827	0.2797	0.4604	24.2
Kuwait	13	0	0	2	2	5	29	0.3320	0.4600	0.7920	24.2
Ukraine	13	0	0	2	3	3	30	0.2560	0.3789	0.75	24.2
Lebanon	13	0	0	2	3	5	4	0.1322	0.3066	0.3966	24.2
Liberia	13	1	2	1	1	4	32	0.0868	0.3066	0.3966	24.2
Malawi	13	0	0	2	3	5	6	0.1216	0.4406	0.5622	24.2
Morocco	13	0	1	1	1	1	3	0.0448	0.3066	0.3514	24.2
Mozambique	13	0	0	2	3	2	9	0.3183	0.3066	0.6249	24.2
Norway	14	0	0	0	0	0	0	0.0000	0.3717	0.3717	24.2
Philippines	14	0	0	2	5	5	48	0.1019	0.2089	0.7600	24.2
Jordan	14	1	0	4	1	1	100	0.5069	0.7099	1.5068	24.2
Romania	14	0	0	3	3	10	6	0.1000	0.3220	0.4220	24.2
Brazil	15	0	0	4	2	9	36	0.5860	0.7700	1.1770	24.2
Russia	15	0	0	3	3	3	11	0.1349	0.3450	0.5449	24.2
Guinea	15	0	0	7	5	3	63	0.2014	0.4534	0.4534	24.2
Malta	15	1	2	2	2	3	7	0.0564	0.8344	0.4040	24.2
India	16	1	0	3	3	3	10	0.2002	0.4642	0.2674	24.2
Kenya	16	0	0	5	3	8	67	0.2002	0.4642	0.2674	24.2
Senegal	16	0	0	5	3	8	67	0.2002	0.4642	0.2674	24.2
Sweden	16	2	0	2	4	4	72	0.6223	0.9103	1.3103	24.2
Romania	16	1	1	3	3	3	9	0.1331	0.4511	1.5211	24.2
Latvia	16	0	0	1	2	1	112	1.3377	1.7877	3.7777	24.2
Madagascar	17	0	0	7	3	7	132	0.4023	1.0343	1.2343	24.2
Maldives	17	0	0	2	2	2	7	0.46	0.46	0.92	24.2
Mexico	19	0	0	3	3	3	11	1.1446	1.1706	2.3146	24.2
Russian Federation	20	0	0	2	3	3	13	0.7119	0.4259	1.2379	24.2
Samoa	20	0	0	2	2	2	10	0.6073	1.015	1.6223	24.2
Dominican Republic	21	0	0	2	2	3	16	0.4009	0.4009	0.8018	24.2
Samoa average	21.48	0.34	0.14	2.64	2.84	1.84	47.0	0.4708	0.6708	1.1408	24.2

(a) Table III: country data, part 3.

Pressure from international donors	Log of the number of deaths caused by accidental poisoning by drugs, including, but not limited to, pesticides, poisons, and liquid household cleaning agents (kg per million inhabitants). Average of the years 1981 through 1998 (the most recent available figure). Source: The United Nations Development Programme (UNDP) (2000).
Deaths from intestinal infection	Log of the number of deaths caused by intestinal infection including digestive disorders per million inhabitants. Average of the years 1981 through 1998 (the most recent available figure). Source: The United Nations Development Programme (UNDP) (2000).
Industrial infrastructure	Population figures are taken from World Bank (2000).
Share of GDP in services	Share of GDP in services (excluding agriculture and manufacturing) as a percentage of total GDP. Source: <i>Author's own calculations</i> (using all countries reported in <i>Indicator and Data</i> (2000) for any given country as well as <i>Sustainable</i> (1996) for Burkina Faso, <i>Chiefs</i> (1996) for the Central African Republic, <i>Tanzania</i> (1996) for Tanzania, and <i>World Development Indicators</i> (1996) for the rest of the world).
Employment in the informal sector	Share of the labor force employed in the informal economy in the capital city of each country as a percentage of the official labor force. <i>Waters</i> are based on the <i>Informal Sector in Latin America and the Caribbean</i> (1996) and <i>Informal Sector in Asia</i> (1996). Source: www.unhcr.org/refugees/industry/industry.htm .
Product market competition	Competition in the product market. Source: <i>Author's own calculations</i> (using the following statement: "Competition in the local market and market share fluctuate constantly." Scale from 1 (strongly disagree) through 7 (strongly agree). Source: <i>IMDI</i> (2002)).
Corruption	Corruption in the business sector. Source: <i>Author's own calculations</i> (using the best available data, e.g., bribery of public officials, kickbacks in public procurement, or embezzlement of public funds). The index averages the corruption scores given by the following sources: (1) <i>Freedom House</i> (1996), (2) <i>Transparency International</i> (1996), (3) <i>World Economic Forum</i> (1996), (4) <i>World Economic Development</i> , Lausanne (IMDI), (5) the International Crime Victim Survey (ICVS) (6) the Political and Economic Risk Observatory, Hong Kong (1996), (7) The Wall Street Journal (1996), (8) <i>World Economic Forum</i> (WEP), December scores from 1 (most corrupt) to 10 (least corrupt). Source: <i>Transparency International</i> (www.transparency.dk).
Gender inequality	Index of "gender gap" in human development. Source: <i>Author's own calculations</i> (using the following statement: "The gender gap is a measure of the difference between 4 = intermediate gender, 5 = substantial inequality, 6 = intermediate gender, 7 = masculine parity or subordination").
Constraints on trade	Index of constraints on the extensive power based on the number of effective steps taken in a country. <i>Waters</i> includes (1) an effective legislative body, (2) an effective executive body, (3) an effective judicial body, (4) an effective administrative body, (5) an effective police force, (6) an effective security force, (7) an effective military force, (8) an effective foreign service, (9) an effective diplomatic corps, (10) an effective intelligence service, (11) an effective information service, (12) an effective communication service, (13) an effective transportation service, (14) an effective postal service, (15) an effective telecommunications service, (16) an effective energy service, (17) an effective water service, (18) an effective health service, (19) an effective education service, (20) an effective social service, (21) an effective cultural service, (22) an effective sports service, (23) an effective entertainment service, (24) an effective tourism service, (25) an effective religious service, (26) an effective legal service, (27) an effective financial service, (28) an effective economic service, (29) an effective industrial service, (30) an effective agricultural service, (31) an effective forestry service, (32) an effective fishing service, (33) an effective hunting service, (34) an effective gaming service, (35) an effective gambling service, (36) an effective lottery service, (37) an effective casino service, (38) an effective hotel service, (39) an effective restaurant service, (40) an effective bar service, (41) an effective club service, (42) an effective lounge service, (43) an effective dance hall service, (44) an effective disco service, (45) an effective nightclub service, (46) an effective theater service, (47) an effective concert service, (48) an effective opera service, (49) an effective ballet service, (50) an effective circus service, (51) an effective puppet service, (52) an effective shadow puppet service, (53) an effective marionette service, (54) an effective acrobatic service, (55) an effective clown service, (56) an effective juggler service, (57) an effective magician service, (58) an effective fortune teller service, (59) an effective astrologer service, (60) an effective palmist service, (61) an effective tarot service, (62) an effective Ouija service, (63) an effective pendulum service, (64) an effective crystal ball service, (65) an effective tea leaf service, (66) an effective coffee service, (67) an effective sugar service, (68) an effective salt service, (69) an effective pepper service, (70) an effective nutmeg service, (71) an effective cloves service, (72) an effective cinnamon service, (73) an effective cardamom service, (74) an effective nutmeg service, (75) an effective pepper service, (76) an effective nutmeg service, (77) an effective pepper service, (78) an effective nutmeg service, (79) an effective pepper service, (80) an effective nutmeg service, (81) an effective pepper service, (82) an effective nutmeg service, (83) an effective pepper service, (84) an effective nutmeg service, (85) an effective pepper service, (86) an effective nutmeg service, (87) an effective pepper service, (88) an effective nutmeg service, (89) an effective pepper service, (90) an effective nutmeg service, (91) an effective pepper service, (92) an effective nutmeg service, (93) an effective pepper service, (94) an effective nutmeg service, (95) an effective pepper service, (96) an effective nutmeg service, (97) an effective pepper service, (98) an effective nutmeg service, (99) an effective pepper service, (100) an effective nutmeg service.
Number of legislature members	Number of members of the legislative body. Source: <i>Author's own calculations</i> (using the following statement: "The number of members of the legislative body is a measure of the size of the legislative body").
Legislature's share of national GDP	Share of national GDP in the legislative sector. Source: <i>Author's own calculations</i> (using the following statement: "The share of national GDP in the legislative sector is a measure of the size of the legislative sector").
Autocracy	Indicates the "general closeness of political institutions" that score from 0 to 10 with 0 being low and 10 being high in autocracy. Average of the years 1981 through 1998 (the most recent available figure). Source: <i>Author's own calculations</i> (using the following statement: "The general closeness of political institutions is a measure of the size of the legislative sector").
Political rights	Index of political rights. Higher index indicates countries that score from 0 to 10 with 0 being low and 10 being high in autocracy. Average of the years 1981 through 1998 (the most recent available figure). Source: <i>Author's own calculations</i> (using the following statement: "The index of political rights is a measure of the size of the legislative sector").

(b) Table II: variables, part 2.

TABLE III

	8	0	0	1	1	6	23	0.1972	0.2892	
Nor Lanka	8	0	0	1	1	5	33	0.1841	0.0081	24.1
Northern Ireland	8	0	1	2	0	5	33	0.1841	0.2313	0.0081
Pakistan, China	8	0	0	1	1	5	37	0.0690	0.0040	13.7
Russia	8	0	0	1	1	5	37	0.1547	0.0040	13.7
Taiwan	8	0	0	2	1	5	50	0.3406	0.0400	4.8
U.S. News	8	0	0	2	1	5	38	0.1966	0.0506	3.7
South Africa	9	0	0	2	2	5	38	0.0844	0.1884	0.0081
Egypt, Republic	9	0	0	1	1	7	32	0.2552	0.3812	0.0081
Thailand	9	0	0	1	1	7	33	0.0629	0.0039	0.0081
Nigeria	9	0	0	1	1	6	36	0.2700	0.2740	0.0081
India	9	0	0	1	1	7	37	0.2773	0.4208	0.0081
Tunisia	9	0	0	0	2	7	41	0.1722	0.3362	0.27
Slovenia	9	0	0	1	1	7	42	0.3303	0.3983	0.0081
Lebanon	9	0	0	1	1	7	63	0.1672	0.1832	0.27
Uganda	10	0	0	1	4	4	23	0.2509	0.0509	0.0081
Bulgaria	10	0	0	2	0	0	27	0.1441	0.0321	0.0081
China	10	0	0	1	1	5	28	0.1308	0.0248	4.1
Germany	10	0	0	1	2	4	42	0.1869	0.0248	25.1
France	10	0	0	1	1	5	45	0.2375	0.3975	0.27
Lithuania	10	0	0	1	1	5	46	0.0646	0.0046	0.0081
Costa Republic	10	0	0	1	2	7	65	0.0822	0.0422	0.6
India	10	0	0	1	2	7	77	0.0562	0.0562	0.0081
Japan	11	0	0	0	2	7	26	0.1361	0.2201	32.2
Canada	11	0	0	1	1	9	39	0.3048	0.0048	0.0081
Egypt, Arab Rep.	11	0	0	1	1	8	51	0.0609	0.1009	1.4
Iran	11	0	0	1	1	9	55	0.0070	0.7230	3.3
Argentina	11	0	0	1	1	9	95	0.1887	0.3487	0.0081
Spain	11	0	0	2	3	1	5	0.2546	0.4006	0.0081
Italy	11	0	0	4	2	1	82	0.1778	0.0010	3.6
Indonesia	11	0	0	3	1	5	108	0.3268	0.1049	0.9
Canada	12	0	0	1	1	9	70	0.0023	0.0023	0.0081
Kazakhstan	12	0	0	1	3	8	42	0.4747	0.4647	1.2
Portugal	12	0	0	1	3	8	78	0.1364	0.0804	0.0081
Slovak Republic	12	0	0	2	3	7	89	0.1452	0.3012	0.8
China	12	0	0	1	1	9	95	0.1887	0.3487	0.0081

(b) Table III: country data, part 2.

TABLE III

Panel B: Means by Quartiles of GDP per capita in 1999											
1 st Quartile	6.77	0.00	0.05	1.58	1.14	4.00	34.50	0.10	0.20	24.873	
2 nd Quartile	11.19	0.24	0.14	2.14	2.38	6.19	49.29	0.33	0.53	34.947	
3 rd Quartile	11.20	0.52	0.14	2.29	2.35	7.14	53.10	0.41	0.52	1.568	
4 th Quartile	11.90	0.62	0.34	2.39	1.85	6.00	63.70	0.80	1.24	840	

Panel C: Test of means (t-statistics)											
1 st vs. 2 nd Quartile	-4.99*	-2.63*	-0.87	-1.35	-3.64*	-3.34*	-3.71*	-0.03*	-0.93*	12.16*	
1 st vs. 3 rd Quartile	-4.58*	-3.04*	-0.87	-1.64*	-2.65*	-4.07*	-4.21*	-2.54*	-1.91*	16.16*	
1 st vs. 4 th Quartile	-4.04*	-2.04*	-0.87	-1.95	-2.45*	-4.07*	-4.09*	-3.57*	-1.60*	17.37*	
2 nd vs. 3 rd Quartile	-1.17	-1.34	0.00	-0.11	0.30	-1.51	-0.54	-0.52	-0.59	6.16	
2 nd vs. 4 th Quartile	-0.33	-0.19	0.00	-0.11	1.91	-0.54	-0.54	-2.54*	-0.79	6.16	
3 rd vs. 4 th Quartile	0.13	-0.27	-0.61	0.11	0.82	0.26	1.06	-0.17	-1.27*	8.48*	

a. Significant at 1 percent; b. significant at 5 percent; c. significant at 10 percent

(b) Table III: country data, part 4.

6.3 Table IV: Evidence on Regulation and Social Outcomes

Read:

- The number of procedures is not associated with more ISO 9000 certifications after income controls.

- More procedures are not associated with lower water pollution or lower deaths from accidental poisoning or intestinal infection.
- More procedures are associated with a larger unofficial economy and lower product-market competition.
- The signs are generally inconsistent with the claim that entry regulation improves observable public goods.

Interpretation: Table IV is the main public-interest test. If regulation screened bad firms and protected consumers, it should show up in quality, health, pollution, or competition. The table finds no such evidence.

Economic message: Entry regulation appears not to buy measurable consumer or public-good benefits; instead, it is associated with informality and weak competition.

TABLE IV
EVIDENCE ON REGULATION AND SOCIAL OUTCOMES

The table presents the results of OLS regressions using the following seven dependent variables: (1) Quality standards as proxied by the number of ISO 9000 certifications; (2) Water pollution; (3) Deaths from accidental poisoning; (4) Deaths from intestinal infection; (5) Size of the unofficial economy as a fraction of GDP; (6) Employment in the unofficial economy; and (7) Product market competition. The independent variables are the log of the number of procedures and the log of per capita GDP in dollars in 1999. Table II describes all variables in detail. Robust standard errors are shown below the coefficients.

Dependent variable	Number of procedures	Ln GDP/POP ₁₉₉₉	Constant	R ² N
Quality standards (ISO Certifications)	-0.2781 ^a (0.0496)		0.7649 ^a (0.1268)	0.3311 85
	-0.1595 ^a (0.0443)	0.0771 ^a (0.0131)	-0.1140 (0.1484)	0.5384 85
Water pollution	0.0127 ^b (0.0084)		0.1557 ^a (0.0174)	0.0247 76
	-0.0037 (0.0076)	-0.0131 ^a (0.0027)	0.2984 ^a (0.0314)	0.2310 76
Deaths from accidental poisoning	0.6588 ^a (0.2057)		1.6357 ^a (0.4381)	0.1179 57
	0.0637 (0.1958)	-0.4525 ^a (0.0933)	6.8347 ^a (1.0929)	0.4109 57
Deaths from intestinal infection	2.3049 ^a (0.3081)		-2.2697 ^a (0.6778)	0.3451 61
	1.0501 ^a (0.2971)	-0.8717 ^a (0.1012)	7.8494 ^a (1.3048)	0.6259 61
Size of the unofficial economy ^d	14.7553 ^a (2.5698)		-3.7982 (5.2139)	0.2482 73
	6.4849 ^b (2.5385)	-6.1908 ^a (1.0834)	67.1030 ^a (13.7059)	0.5187 73
Employment in the unofficial economy	19.4438 ^a (2.5756)		-4.1103 (5.9160)	0.3132 46
	13.8512 ^a -3.6056	-4.4585 ^a (1.3918)	41.5133 ^b (17.6836)	0.4477 46
Product market competition	-0.4012 ^a (0.1213)		5.7571 ^a (0.2511)	0.1405 54
	-0.1418 (0.1202)	0.2108 ^a (0.0680)	3.3579 ^a (0.7749)	0.3087 54

Table IV: evidence on regulation and social outcomes.

6.4 Figure III: Corruption and Number of Procedures and Table V: Evidence on the Tollbooth Theory

Read:

- Figure III shows a strong negative relation between the corruption index and log number of procedures: countries with more procedures are more corrupt.
- Table V estimates this relationship using number of procedures, time, and cost.
- The number-of-procedures coefficient is negative and highly significant with and without GDP per capita controls.
- The association remains when the sample is split into above- and below-median income countries.

Interpretation: This is the paper's central public-choice evidence. Regulation creates many points at which officials can extract payments. The association with corruption is large and robust.

Economic message: The tollbooth analogy fits the data: more tollbooths in the entry process are associated with more corruption, not better regulation.

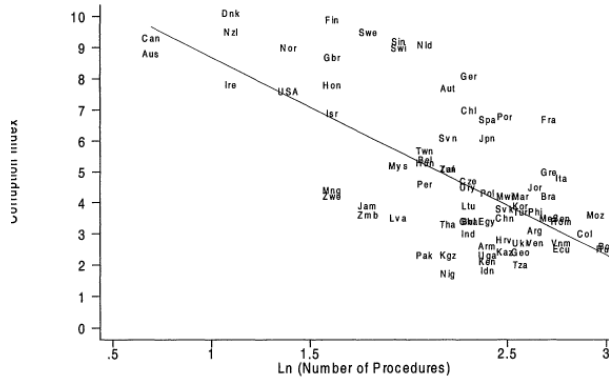


FIGURE III
Corruption and Number of Procedures
The scatter plot shows the values of the corruption index against the number of procedures for the 78 countries in our sample with nonmissing data.

(a) Figure III: corruption and number of procedures.

TABLE V
EVIDENCE ON THE TOLLBOOTH THEORY
The table presents the results of OLS regressions using corruption as the dependent variable. The independent variables are (1) the log of the number of procedures; (2) the log of time; (3) the log of cost; and the log of per capita GDP in dollars in 1999. Panel A presents results for the 78 observations with available corruption data. Panel B reports results separately for the subsample of countries with GDP per capita in 1999 above and below the sample median. Table II describes all variables in detail. Robust standard errors are shown in parentheses below the coefficients.

Panel A: Results for the whole sample						
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)
Number of procedures	-3.1811 ^a (0.2986)	-1.8654 ^a (0.2131)				
Time			-1.7566 ^a (0.1488)	-0.8854 ^a (0.1377)		
Cost					-1.2129 ^a (0.1206)	-0.4978 ^a (0.1285)
Ln GDP/POP ₁₉₉₉		0.9966 ^a (0.0864)		0.9765 ^a (0.1014)		0.9960 ^a (0.1118)
Constant	11.8741 ^a (0.7380)	1.1345 (0.9299)	11.0694 ^a (0.5932)	0.0677 (1.1176)	2.7520 ^a (0.2414)	-4.0893 ^a (0.7867)
R ²	0.4656	0.8125	0.4387	0.7662	0.4256	0.7306
N	78	78	78	78	78	78
Panel B: Results for countries above and below the world median GDP per capita						
	Countries above median GDP/POP ₁₉₉₉			Countries below median GDP/POP ₁₉₉₉		
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)
Number of procedures	-1.8729 ^a (0.2971)			-0.7841 ^b (0.3304)		
Time		-0.8135 ^a (0.1762)			-0.0923 (0.2850)	
Cost			-0.5327 ^a (0.1894)			-0.3408 ^a (0.1021)
Ln GDP/POP ₁₉₉₉	1.4811 ^a (0.2265)	1.5871 ^a (0.2789)	1.7621 ^a (0.2913)	0.3993 ^b (0.1735)	0.3680 ^a (0.1802)	0.2117 (0.1718)
Constant	-3.6970 (2.4628)	-5.9027 ^c (2.9942)	-11.3736 ^a (2.5773)	2.3246 ^c (1.2849)	1.0098 (1.8813)	1.3125 (1.1136)
R ²	0.7820	0.7155	0.6728	0.2362	0.1324	0.2830
N	40	40	40	38	38	38

(b) Table V: evidence on the tollbooth theory.

6.5 Table VI: Correlation Table for Political Attributes

Read:

- Democratic and limited-government variables are strongly correlated with each other.
- More democratic institutions correlate negatively with regulation measures.

- Legal origin is less strongly correlated with the political variables, except for socialist legal origin.
- Income is positively associated with democracy and limited government.

Interpretation: Table VI is not an outcome regression. It shows the structure of institutional variables and warns that political variables are highly collinear. Therefore, Table VII estimates them separately.

Economic message: Regulation, political institutions, and income are deeply linked; separating their effects is hard, but the correlation structure itself is informative.

TABLE VI
CORRELATION TABLE FOR POLITICAL ATTRIBUTES
The table reports correlations among measures of regulation and the variables used in Table VII. All variables are defined in Table II. Significance levels are Bonferroni-adjusted.

	Exec de facto independence	Constraints on executive power	Effectiveness legislature	Competition nominating	Autocracy	Political rights	French LO
Exec de facto independence	1.0000						
Constraints on exec. power	0.9761 ^a	1.0000					
Effectiveness legislature	0.9210 ^a	0.9078 ^a	1.0000				
Competition nominating	0.8243 ^a	0.8069 ^a	0.8484 ^a	1.0000			
Autocracy	-0.9085 ^a	-0.8844 ^a	-0.8514 ^a	-0.7819 ^a	1.0000		
Political rights	0.8440 ^a	0.8448 ^a	0.8485 ^a	0.7191 ^a	-0.8564 ^a	1.0000	
French legal origin	-0.1814	-0.1814	-0.1901	-0.1985	-0.0258	0.0565	1.0000
Socialist legal origin	-0.3321	-0.2927	-0.3236	-0.3240	0.5475 ^a	-0.4572 ^a	-0.4169 ^a
German legal origin	0.2101	0.2008	0.2023	0.1281	-0.1920	0.2444	-0.2141
Scandinavian legal origin	0.3391	0.3274	0.3378	0.2522	-0.2978	0.3109	-0.1727
English legal origin	0.2259	0.1998	0.1462	0.2412	-0.2324	0.0778	-0.4874 ^a
Ln GDP/POP ₁₉₉₉	0.6900 ^a	0.6703 ^a	0.7483 ^a	0.6123 ^a	-0.6389 ^a	0.7519 ^a	-0.0767 ^b
Ln(Number of procedures)	-0.5518 ^a	-0.5234 ^a	-0.5848 ^a	-0.4435 ^b	0.4662 ^a	-0.4412 ^a	0.4863 ^a
Ln(Time)	-0.5420 ^a	-0.5204 ^a	-0.5635 ^a	-0.4360 ^b	0.4770 ^a	-0.4921 ^a	0.3976 ^b
Ln(Cost)	-0.5070 ^a	-0.4937 ^a	-0.5656 ^a	-0.4177 ^b	0.4075 ^b	-0.4588 ^a	0.3472
Ln(Cost + time)	-0.5700 ^a	-0.5478 ^a	-0.6267 ^a	-0.4745 ^a	0.4713 ^a	-0.5085 ^a	0.3870 ^b

(a) Table VI: political correlations, part 1.

TABLE VI
(CONTINUED)

Socialist LO	German LO	Scandinavian LO	English LO	Ln GDP/POP ₁₉₉₉	Ln (Number of procedures)	Ln (Time)	Ln (Cost)	Ln (Cost + time)
1.0000								
-0.1479	1.0000							
-0.1192	-0.0612	1.0000						

(b) Table VI: political correlations, part 2.

6.6 Table VII: Evidence on Regulation and Political Attributes and Figure IV: Autocracy and Number of Procedures

Read:

- Table VII regresses log number of procedures on political and legal-origin variables, controlling for log GDP per capita.
- More independent executives, stronger executive constraints, more effective legislatures, and more political rights are associated with fewer procedures.
- Autocracy is positively associated with regulation.
- French, German, and Socialist legal origins are associated with more procedures than English origin.
- Figure IV shows the positive relationship between autocracy and log number of procedures.

Interpretation: Table VII is the main political-economy test. The public-interest view predicts more representative governments should regulate more if regulation protects consumers. The data show the opposite.

Economic message: Entry regulation is heavier where public oversight is weaker and the state is more interventionist, consistent with rent extraction by politicians and bureaucrats.

TABLE VII
EVIDENCE ON REGULATION AND POLITICAL ATTRIBUTES

The table presents the results of running regressions for the log of the number of procedures as the dependent variable. We run seven regressions using various political indicators described in Table II and (log) GDP per capita. Robust standard errors are shown in parentheses below the coefficients.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Executive de facto independence	-0.1249* (0.0322)						
Constraints on executive power		-0.1048* (0.0352)					
Effectiveness of legislature			-0.3391* (0.0778)				
Competition nominating				-0.2736* (0.0999)			
Autocracy					0.0043* (0.0170)		
Political rights						-0.3470 (0.2185)	
French legal origin							0.7343* (0.0914)
Socialist legal origin							0.4904* (0.1071)
German legal origin							0.7759* (0.1363)
Scandinavian legal origin							-0.0988 (0.1733)
Ln GDPPOP ₁₉₉₅	-0.0491 (0.0331)	-0.0634* (0.0352)	-0.0087 (0.0403)	-0.0002* (0.0353)	-0.0897* (0.0321)	-0.0809* (0.0360)	-0.1434* (0.0270)
Constant	3.1782* (0.2234)	3.2600* (0.2408)	2.8709* (0.2346)	3.3540* (0.2841)	2.7457* (0.2385)	3.1850* (0.2399)	2.9492* (0.2650)
R ²	0.1778	0.1972	0.3424	0.2473	0.2640	0.2350	0.4558
N	84	84	73	73	84	84	85

a. Significant at 1 percent; b. significant at 5 percent; c. significant at 10 percent.

(a) Table VII: regulation and political attributes.

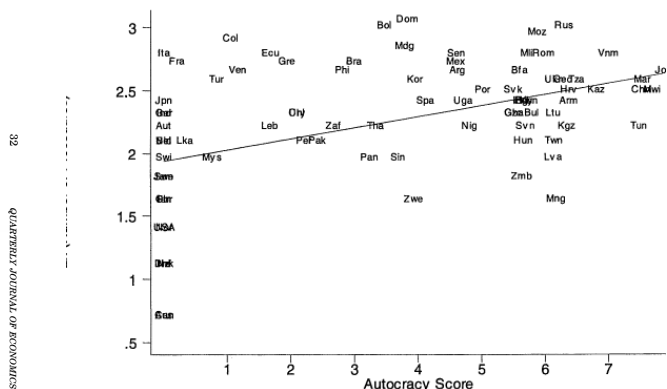


FIGURE IV
Autocracy and Number of Procedures

The scatter plot shows the values of the (log) number of procedures against autocracy score (higher values for more autocratic systems) for the 84 countries.

(b) Figure IV: autocracy and number of procedures.

7 Interpretive synthesis

- **Public-interest theory fails the outcome test.** Regulation is not associated with better quality, pollution, health, or competition outcomes.
- **Public-choice theory passes the rent-extraction test.** More regulation is associated with corruption and a larger unofficial economy.
- **Political institutions reinforce the interpretation.** Less constrained and less democratic governments regulate more heavily.
- **Regulation is not only a cost.** It also changes the allocation of economic activity: when formal entry is costly, entrepreneurs may remain informal or not enter at all.
- **Legal origin matters, but differently.** Civil-law and socialist legal origins are associated with more entry procedures, consistent with a more interventionist state tradition.

8 Short summary

- The paper builds a new dataset on official entry procedures, time, and cost for 85 countries.
- Entry regulation varies enormously across countries.
- Rich countries generally have fewer procedures, shorter official times, and lower official costs.
- Heavy regulation is not associated with better public-good outcomes.
- Heavy regulation is associated with more corruption and larger informal economies.
- Democratic, limited governments regulate entry less, not more.
- Autocracy and civil-law/socialist legal origins are associated with heavier regulation.
- The evidence supports public-choice and tollbooth theories more than public-interest theories.

9 Conclusion

9.1 Core conclusion

Djankov, La Porta, Lopez-de-Silanes, and Shleifer show that formal business-entry regulation is extremely burdensome in many countries and that heavier regulation is associated with corruption and informality rather than improved social outcomes. The evidence supports the view that entry regulation often creates rents for politicians and bureaucrats.

9.2 Economic intuition

If entry regulation protected consumers, countries with more procedures should have better product quality, cleaner environments, better health outcomes, or more competition. Instead, they have more corruption and larger unofficial sectors. Regulation acts like a tollbooth: every procedure creates a gate at which officials can extract rents.

9.3 Policy takeaway

Simplifying legal entry procedures can reduce opportunities for corruption and encourage formal entrepreneurship. The policy lesson is not that all regulation is bad, but that entry regulation should be evaluated by its actual social benefits, not by its stated intention to protect consumers.

9.4 Incremental contribution in one sentence

The paper transformed the regulation-of-entry debate by introducing a comparable cross-country dataset and using it to show that formal entry barriers are more consistent with rent extraction than with public-interest protection.